

Advanced Methodologies for House Price Prediction

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I. Introduction

Understanding house prices is crucial for various stakeholders in the real estate industry, including homeowners, potential buyers, investors, real estate agents, and policy makers. Housing is often the largest investment people make, and fluctuations in house prices have significant economic implications. Accurate price prediction models can help buyers and sellers make informed decisions, help banks and financial institutions in risk assessment, and provide information for government policy related to housing affordability and real estate taxation.

House prices are influenced by a wide range of factors, including local market conditions, property attributes, and economic indicators. Studying these prices helps to understand the factors that drive the housing market, enabling more transparent and data-driven decisions. Furthermore, recent advances in machine learning techniques provide new ways to improve the accuracy of house price prediction by leveraging complex patterns and relationships within the data. The insights gained from this analysis could be invaluable for various stakeholders, including homeowners, investors, and policymakers, making this project both academically interesting and practically impactful.

II. Data

The data for this project come from the House Prices - Advanced Regression Techniques competition on Kaggle. This data set includes information on residential properties in Ames, Iowa, collected by Dean De Cock for educational purposes. The data set contains a training set of 1,460 observations and 79 explanatory variables that describe aspects of residential homes, together with the sale price as the target variable. A test set without target values is also available for model evaluation.

The data set includes a diverse array of variables that span various aspects of a property, including: Property-specific features: Lot size, year built, total number of rooms, overall quality, and condition ratings. Neighborhood-specific features: Location-based information,

such as neighborhood and proximity to key amenities. Structural details: Basement square footage, garage area, type of heating and exterior materials. Sales conditions: Timing of sale and any conditions that affect the sale, such as partial or full sale. This data provides a comprehensive view of property features, allowing for complex modeling and predictions of housing prices. Given the high-dimensional and varied nature of the features, this dataset presents a valuable opportunity to apply advanced statistical and machine learning techniques to extract meaningful insights and create an accurate pricing model. The following is a summary of some key features.

The mean lot size for a house is approximately 10,516 square feet, with a standard deviation of 9,981 square feet, indicating significant variability in lot sizes. The smallest lot is 1,300 square feet, while the largest spans a massive 215,245 square feet. Also, the overall quality of the houses has an average score of 6.1 on a scale of 1 to 10, with a standard deviation of 1.4. The range is from the lowest quality score of 1 to the highest quality score of 10. The average construction year is approximately 1971, with houses ranging from as old as 1872 to as recent as 2010. The standard deviation of 30 years reflects a broad spread in construction periods.

The mean above-ground living area is 1,515 square feet, with a standard deviation of 525 square feet. The smallest recorded living area is 334 square feet, and the largest is 5,642 square feet, indicating a wide range in home sizes. On average, houses have space for 1.8 cars in the garage, with a standard deviation of 0.7 cars. Some houses lack garages (0 cars), while the largest garage can accommodate up to 4 cars. The mean number of fireplaces is 0.6, showing that many houses may have no fireplaces, as reflected by a minimum of 0 and a maximum of 3 fireplaces. The standard deviation of 0.6 suggests moderate variability in this feature. The average wood deck area is 94 square feet, with a standard deviation of 125 square feet. Some houses have no wood decks (0 square feet), while the largest deck is 857 square feet. The average sale price of the houses is \$180,921, with a standard deviation of \$79,442, indicating considerable variation in home prices. Prices range from a minimum of \$34,900 to a maximum of \$755,000, covering a broad spectrum of affordability and luxury.

Table 1—Summary Statistics

	Mean	SDV	Minimum	Maximum
MSSubClass	56.897	42.301	20.000	190.000
LotArea	10516.828	9981.265	1300.000	215245.000
OverallQual	6.099	1.383	1.000	10.000
OverallCond	5.575	1.113	1.000	9.000
YearBuilt	1971.268	30.203	1872.000	2010.000
YearRemodAdd	1984.866	20.645	1950.000	2010.000
BsmtFinSF1	443.640	456.098	0.000	5644.000
BsmtFinSF2	46.549	161.319	0.000	1474.000
BsmtUnfSF	567.240	441.867	0.000	2336.000
TotalBsmtSF	1057.429	438.705	0.000	6110.000
1stFlrSF	1162.627	386.588	334.000	4692.000
2ndFlrSF	346.992	436.528	0.000	2065.000
LowQualFinSF	5.845	48.623	0.000	572.000
GrLivArea	1515.464	525.480	334.000	5642.000
BsmtFullBath	0.425	0.519	0.000	3.000
BsmtHalfBath	0.058	0.239	0.000	2.000
FullBath	1.565	0.551	0.000	3.000
HalfBath	0.383	0.503	0.000	2.000
BedroomAbvGr	2.866	0.816	0.000	8.000
KitchenAbvGr	1.047	0.220	0.000	3.000
TotRmsAbvGrd	6.518	1.625	2.000	14.000
Fireplaces	0.613	0.645	0.000	3.000
GarageCars	1.767	0.747	0.000	4.000
GarageArea	472.980	213.805	0.000	1418.000
WoodDeckSF	94.245	125.339	0.000	857.000
OpenPorchSF	46.660	66.256	0.000	547.000
EnclosedPorch	21.954	61.119	0.000	552.000
3SsnPorch	3.410	29.317	0.000	508.000
ScreenPorch	15.061	55.757	0.000	480.000
PoolArea	2.759	40.177	0.000	738.000
MiscVal	43.489	496.123	0.000	15500.000
MoSold	6.322	2.704	1.000	12.000
YrSold	2007.816	1.328	2006.000	2010.000
SalePrice	180921.196	79442.503	34900.000	755000.000

III. Methodology

To effectively model house prices using the dataset, I propose to explore a range of advanced machine learning methodologies, specifically focusing on methods that can capture nonlinear relationships and interactions between variables.

1) Linear Models

In this study, we employ a Generalized Linear Model (GLM) as a baseline model for predicting house prices. GLM is an extension of the ordinary linear regression model that allows the dependent variable (or response) to follow distributions other than the normal distribution. GLMs are widely used in statistical modeling to handle cases where the assumptions of linear regression (e.g., normality and constant variance) do not hold.

2) Tree-Based Models

Tree-based models are powerful for predictive tasks because they handle nonlinear relationships well and can manage high-dimensional data. In this project, we propose to use the following tree-based methods:

- a) Random Forests: Random Forests are an ensemble learning method that constructs multiple decision trees and aggregates their results to improve model robustness and reduce overfitting (Segal, 2004). This approach is highly interpretable, as we can assess the importance of each variable in predicting house prices, helping us understand the drivers of pricing in the dataset. Bagging is a special case of Random Forest where all predictors are considered at each split (Breiman, 1996).
- b) Gradient Boosting Machines (GBM): GBMs iteratively build decision trees to correct errors made by previous models, allowing for fine-tuning and greater accuracy. Variants of GBMs, such as XGBoost and LightGBM, are computationally efficient and often produce state-of-the-art results for structured data tasks (Ke et al., 2017). These models allow for tuning of learning rate, tree depth, and number of estimators, providing flexibility in capturing complex relationships in the data.

- c) Bayesian Additive Regression Trees (BART): BART provide a probabilistic approach to regression, combining the flexibility of tree-based models with Bayesian inference. BART models predict the response variable by summing the contributions of multiple small trees, each representing a different feature interaction. According to Chipman et al. (2010), this approach allows for a robust prediction interval, giving insight into the uncertainty of predictions—an advantage over traditional tree-based models. BART is especially suitable for this dataset because it can model complex, non-linear interactions without extensive feature engineering. Moreover, its Bayesian nature allows it to account for uncertainty in predictions, which is valuable for financial forecasting, as it enables stakeholders to evaluate risk alongside the predicted prices.

Tree-based models are advantageous for this dataset due to their ability to automatically handle categorical features which are common in real-world data. Additionally, they are resilient to multicollinearity, which is often present in high-dimensional data.

3) Generalized Additive Models (GAM)

Generalized Additive Models (GAM) are an extension of traditional linear models that allow for non-linear relationships between the response and predictor variables by applying smooth functions to each variable (Hastie, 2017). GAMs provide a balance between interpretability and flexibility, as they maintain additive structure while allowing individual predictors to have non-linear effects. This can be beneficial for house price prediction, where certain features (e.g., year built, lot size) may have a non-linear influence on price. GAMs are advantageous for interpretability, as they clearly show the effect of each feature on the response variable, which is crucial for explaining predictions to stakeholders. Additionally, GAMs can reveal insights into the influence of certain property features on price, enabling data-driven decision-making for buyers and sellers.

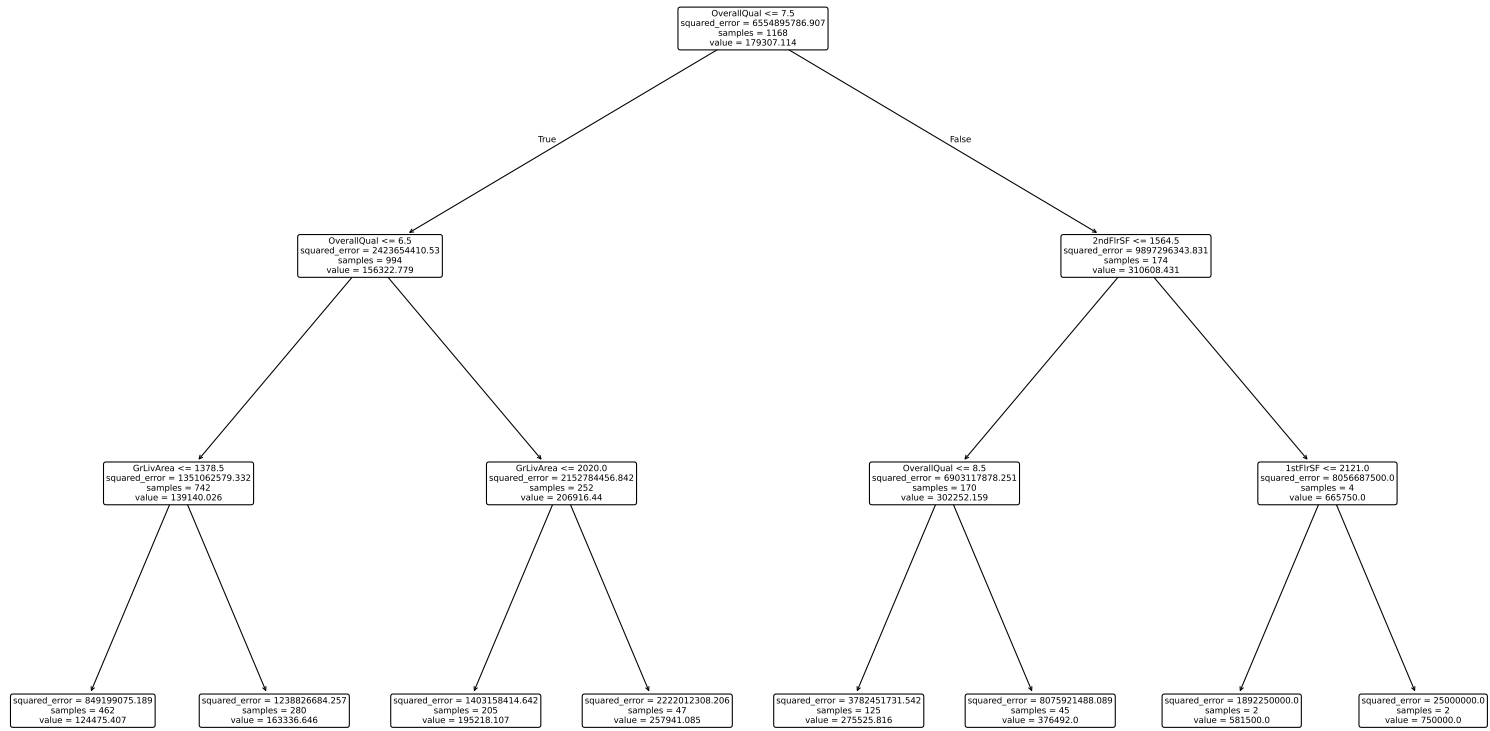
IV. Results

A. Generalized Linear Models (GLM)

We first explore the performance of a linear model which will serve as a baseline for measuring the predictive performance of the subsequent models. I used Gamma as my target distribution with a log link. From the result in Table 3 below, we see that some of the significant features in OverallQual, GRLivArea, TotalBsmtSF, 2ndFlrSF, and 1stFlrSF.

B. Tree-based Models

Figure 1. Decision Tree (Max Depth = 3)



We start off with a single tree grown with a maximum depth of 3. Each terminal node represents a subset of houses that share similar characteristics as determined by the splits. Below is an explanation of the path to each terminal node:

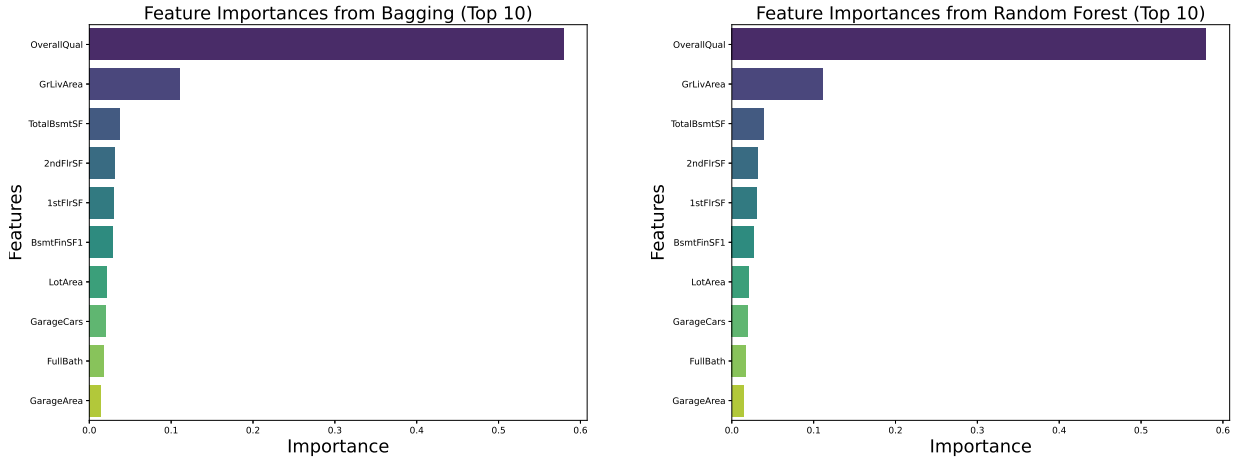
A house with $\text{OverallQual} \leq 6.5$, and $\text{GrLivArea} \leq 1378.5$ is predicted to have a sale price of \$124,475. These houses have lower overall quality and smaller living areas. A house with $\text{OverallQual} \leq 6.5$ and GrLivArea between 1378.5 and ≤ 2020.0 is predicted to have a price of \$163,336. These houses have lower overall quality but moderately sized living areas. A house with OverallQual between 6.5 and 7.5 and $\text{GrLivArea} \leq 2020.0$ is predicted to have a price of \$195,218. These houses have lower overall quality but larger living areas. A house with OverallQual between 6.5 and 7.5 and $\text{GrLivArea} > 2020.0$ is predicted to have a price of \$257,941. These houses have higher overall quality.

A house with OverallQual between 7.5 and 8.5 and $\text{2ndFlrSF} \leq 1564.5$ is predicted to have a price of \$275,525. These houses have higher overall quality and larger second floors. A house with $\text{OverallQual} > 8.5$ and $\text{2ndFlrSF} \leq 1564.5$ is predicted to have a price of \$376,492. A house with $\text{OverallQual} > 7.5$, $\text{2ndFlrSF} > 1564.5$, and $\text{1stFlrSF} \leq 2121.0$ is predicted to have a price of \$581,500. These houses have the highest overall quality and moderately sized first floors. A house with $\text{OverallQual} > 7.5$, $\text{2ndFlrSF} > 1564.5$, and $\text{1stFlrSF} > 2121.0$ is predicted to have a price of \$750,000. These houses have the highest overall quality and the largest first floors.

I also looked at the performance of two ensemble trees, namely; Bagging and Random forest. To assess the contribution of each feature on the models, we analyzed the feature importance graph which is presented in Figure 2 below. In Bagging, feature importance is often derived by measuring the reduction in a loss function (such as Mean Squared Error for regression tasks) when the variable is used for a split in the tree. In Random Forests, variable importance is computed importance of a variable is calculated as the average reduction in loss across all trees in the ensemble. In Random Forests, variable importance is computed as the total decrease in impurity (variance reduction in regression) when a variable is used in splits across all trees. For both models, the most important predictor of House Price

is the Overall Quality of the house, followed by GrLivArea, TotalBsmtSF, 2ndFlrSF, and 1stFlrSF. This means that the overall quality of the house is the most significant predictor of house price which makes intuitive sense since it encapsulate almost all the other features.

Figure 2. Feature Importance Graphs

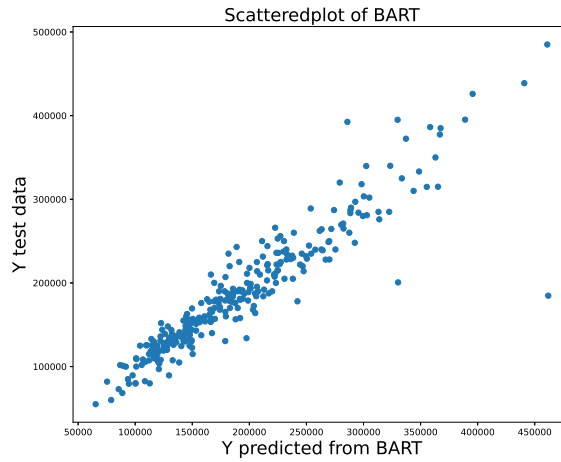


I also looked at other modern tree-based approach to modeling this problem including gradient boosting regression tree, XGBoost and Light XGBoost. The results of this will be discussed in details in the concluding section.

C. Bayesian Additive Regression Tree (BART)

As mentioned earlier, the BART approach allows for a robust prediction interval, giving insight into the uncertainty of predictions—an advantage over traditional tree-based models. From Figure 3, we observe an almost perfect positive linear correlation between the predicted Sale Prices from the BART model and the actual Sale Prices in the test dataset. This is an indication the model does fit the data well.

Figure 3. BART Scatter plot



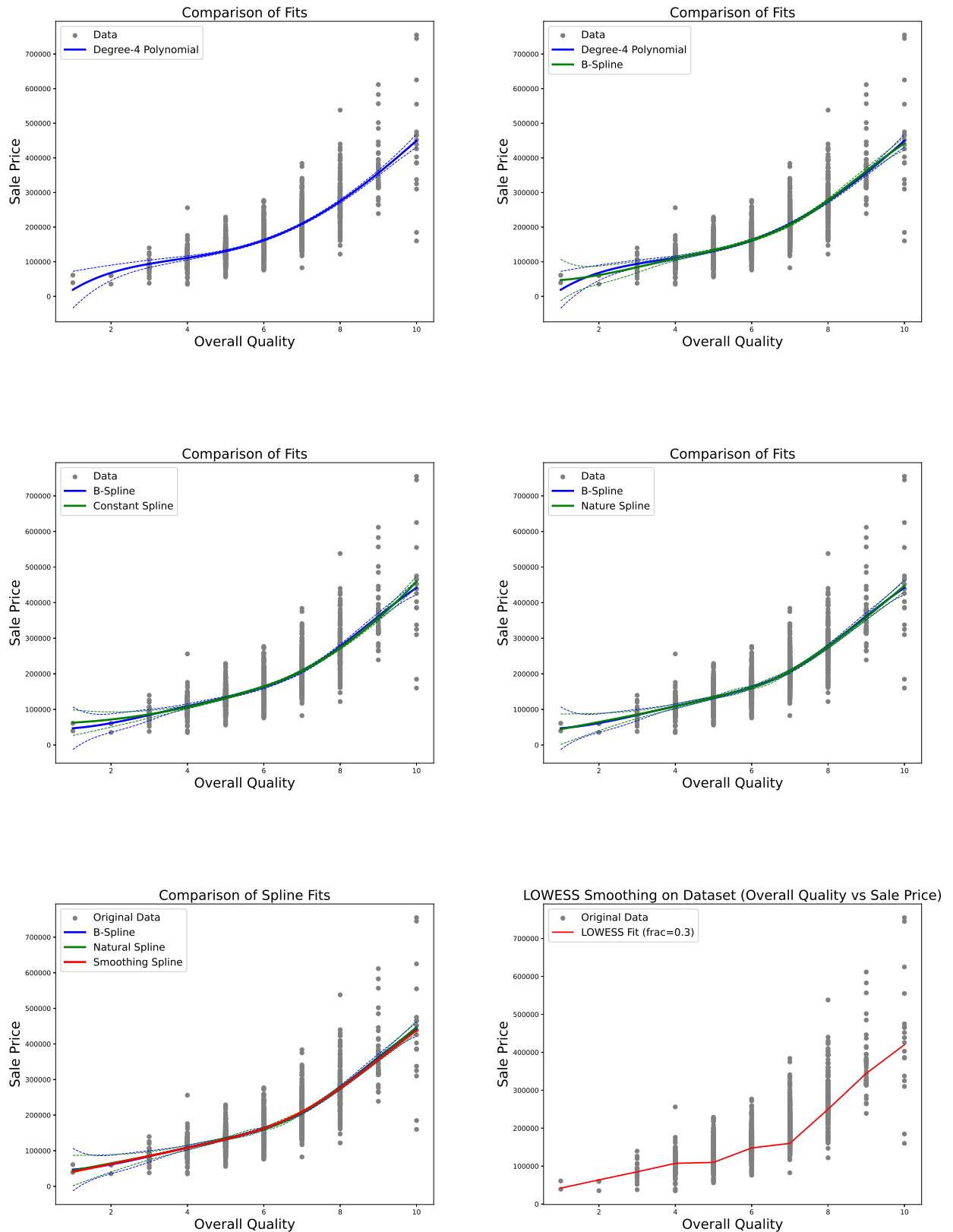
D. Generalized Additive Models (GAM)

The results shown in the figure provide a detailed comparison of various approaches used to fit a Generalized Additive Model (GAM) to predict house prices based on Overall Quality. The Degree-4 Polynomial Fit provides a smooth, continuous curve to model the relationship between Overall Quality and Sale Price. It captures the general trend but may overfit the data in certain regions due to its global nature. The B-Spline Fit introduces more flexibility by dividing the data into smaller segments and fitting piecewise polynomials. This allows it to better adapt to localized patterns in the data while avoiding overfitting in areas with less variation. The Constant Spline assumes a stepwise relationship between Overall Quality and Sale Price, where changes occur in discrete steps. This fit is less smooth and might oversimplify the data, missing subtle variations. The Natural Spline provides a piecewise smooth fit with boundary constraints. This ensures a more realistic representation of the relationship while maintaining smooth transitions at the edges.

All models reveal a strong positive relationship between Overall Quality and Sale Price, indicating that higher quality homes command higher prices. However, the degree of flexibility in capturing this relationship varies across methods. For global trends and interpretability, Degree-4 Polynomial might be preferred. For localized patterns and flexibility, B-Spline or

Natural Spline works well. LOWESS is suitable for purely exploratory analyses, providing robust insights into local variations. Smoothing Spline or Natural Spline appears to strike a balance between capturing overall trends and local nuances while avoiding overfitting. When interpretability is crucial, simpler models like Degree-4 Polynomial can be considered. For further prediction tasks, model validation and comparison on unseen data should guide the final choice.

Figure 4. Comparison of Different GAM Fits



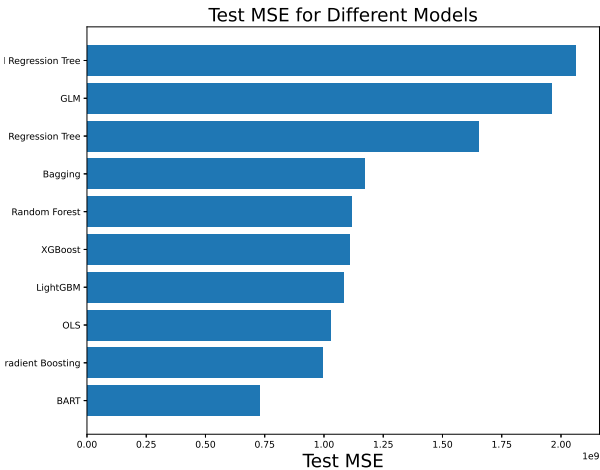
V. Conclusion

The Kaggle House Prices dataset offers a rich source of information for building a robust and interpretable model for predicting house prices. By leveraging advanced methodologies such as Random Forests, Gradient Boosting, BART, and GAM, we can effectively handle the dataset’s complexity and improve prediction accuracy. The summary of the models and their respective MSE are presented in the table below.

Table 2—Test MSE for Different Models

Model	Test MSE
OLS	1027833125.3322679
GLM	1960774876.6565
Regression Tree	1651040862.408
Pruned Regression Tree	2059448737.675
Bagging	1170152884.750
Random Forest	1114393872.969
Gradient Boosting	994793928.2664747
XGBoost	1107646740.3252983
LightGBM	1083696710.0096252
BART	729837253.536

Figure 5. Model Comparison Plot



From the results, BART demonstrates the best predictive accuracy with the lowest test MSE (729,837,253.54). This suggests that BART is well-suited for capturing the complex relationships in the data. Among tree-based ensemble methods, Gradient Boosting performs admirably (994,793,928.27), followed by Random Forest (1,114,393,872.97) and Bagging (1,170,152,884.75). These methods benefit from combining multiple trees to reduce variance and bias. Other models such as OLS (1,027,833,125.33) and GLM (1,960,774,876.66) show higher MSE values, indicating they are less effective at modeling non-linear relationships and interactions. Similarly, Pruned Regression Tree and Regression Tree have test MSEs of 2,059,448,737.68 and 1,651,040,862.41, respectively, reflecting their limitations as single-tree models.

Based on this analysis, BART is the recommended model due to its superior predictive accuracy. However, Gradient Boosting could be a strong alternative, especially in scenarios where interpretability and computational efficiency are prioritized.

References

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Appendix

Table 3—Generalized Linear Model

Dependent variable: Sale Price	
1stFlrSF	0.000*** (0.000)
2ndFlrSF	0.000** (0.000)
3SsnPorch	0.000 (0.000)
BedroomAbvGr	-0.002 (0.007)
BldgType__2fmCon	0.043 (0.058)
BldgType__Duplex	-0.038 (0.034)
BldgType__Twnhs	-0.052 (0.048)
BldgType__TwnhsE	-0.018 (0.043)
BsmtFinSF1	0.000*** (0.000)
BsmtFinSF2	0.000 (0.000)
BsmtFullBath	0.016* (0.010)
BsmtHalfBath	-0.007 (0.014)
BsmtUnfSF	-0.000 (0.000)
CentralAir__Y	0.052*** (0.018)
Condition1__Feedr	0.006 (0.025)
Condition1__Norm	0.051** (0.020)
Condition1__PosA	0.010 (0.052)
Condition1__PosN	0.034 (0.036)
Condition1__RRAe	-0.043 (0.045)
Condition1__RRNn	0.063 (0.066)
Condition1__RRNe	-0.022 (0.079)
Condition1__RRAn	0.029 (0.035)

Dependent variable: SalePrice	
Condition2_Feedr	0.160 (0.105)
Condition2_Norm	0.093 (0.086)
Condition2_PosA	0.159 (0.166)
Condition2_PosN	-0.046 (0.140)
Condition2_RRAe	-0.415 (0.276)
Condition2_RRAn	0.032 (0.137)
Condition2_RRNn	-0.102 (0.140)
EnclosedPorch	0.000 (0.000)
ExterCond_Fa	-0.114 (0.082)
ExterCond_Gd	-0.091 (0.078)
ExterCond_Po	-0.198 (0.142)
ExterCond_TA	-0.078 (0.078)
ExterQual_Fa	0.035 (0.051)
ExterQual_Gd	0.007 (0.023)
ExterQual_TA	0.008 (0.026)
Exterior1st_AsphShn	0.038 (0.150)
Exterior1st_BrkComm	-0.151 (0.139)
Exterior1st_BrkFace	0.153** (0.063)
Exterior1st_CBlock	-0.033 (0.059)
Exterior1st_CemntBd	-0.067 (0.101)
Exterior1st_HdBoard	0.080 (0.064)
Exterior1st_ImStucc	0.055 (0.126)
Exterior1st_MetalSd	0.154** (0.072)

Dependent variable: SalePrice	
Exterior1st_Plywood	0.058 (0.063)
Exterior1st_Stone	0.089 (0.139)
Exterior1st_Stucco	0.089 (0.067)
Exterior1st_VinylSd	0.095 (0.070)
Exterior1st_Wd Sdng	0.066 (0.063)
Exterior1st_WdShing	0.075 (0.066)
Exterior2nd_AsphShn	-0.002 (0.106)
Exterior2nd_Brk Cmn	0.000 (0.111)
Exterior2nd_BrkFace	-0.100 (0.069)
Exterior2nd_CBlock	-0.033 (0.059)
Exterior2nd_CmentBd	0.107 (0.101)
Exterior2nd_HdBoard	-0.062 (0.066)
Exterior2nd_ImStucc	-0.035 (0.073)
Exterior2nd_MetalSd	-0.101 (0.073)
Exterior2nd_Other	-0.167 (0.127)
Exterior2nd_Plywood	-0.044 (0.064)
Exterior2nd_Stone	-0.059 (0.090)
Exterior2nd_Stucco	-0.043 (0.068)
Exterior2nd_VinylSd	-0.058 (0.071)
Exterior2nd_Wd Sdng	-0.040 (0.064)
Exterior2nd_Wd Shng	-0.063 (0.066)
Fireplaces	0.020*** (0.006)
Foundation_CBlock	0.015 (0.015)
Foundation_PConc	0.036** (0.017)

Dependent variable: SalePrice	
Foundation__Slab	-0.022 (0.038)
Foundation__Stone	0.163*** (0.058)
Foundation__Wood	-0.095 (0.079)
FullBath	0.016 (0.011)
Functional__Maj2	-0.241*** (0.060)
Functional__Min1	0.003 (0.040)
Functional__Min2	-0.006 (0.040)
Functional__Mod	-0.099* (0.051)
Functional__Sev	-0.413*** (0.118)
Functional__Typ	0.044 (0.035)
GarageArea	0.000*** (0.000)
GarageCars	0.032*** (0.011)
GrLivArea	0.000*** (0.000)
HalfBath	0.019* (0.010)
HeatingQC__Fa	-0.032 (0.022)
HeatingQC__Gd	-0.025** (0.010)
HeatingQC__Po	-0.167 (0.119)
HeatingQC__TA	-0.043*** (0.010)
Heating__GasA	0.133 (0.108)
Heating__GasW	0.154 (0.113)
Heating__Grav	-0.036 (0.117)
Heating__OthW	0.074 (0.135)
Heating__Wall	0.248** (0.126)
HouseStyle__1.5Unf	0.030 (0.037)

Dependent variable: SalePrice	
HouseStyle__1Story	-0.027 (0.021)
HouseStyle__2.5Fin	-0.078 (0.058)
HouseStyle__2.5Unf	0.059 (0.048)
HouseStyle__2Story	-0.009 (0.017)
HouseStyle__SFoyer	0.011 (0.029)
HouseStyle__SLvl	0.022 (0.026)
KitchenAbvGr	-0.044 (0.028)
KitchenQual__Fa	-0.044 (0.030)
KitchenQual__Gd	-0.080*** (0.017)
KitchenQual__TA	-0.078*** (0.019)
LandContour__HLS	0.029 (0.026)
LandContour__Low	0.002 (0.031)
LandContour__Lvl	0.017 (0.018)
LandSlope__Mod	0.030 (0.019)
LandSlope__Sev	-0.228*** (0.055)
LotArea	0.000*** (0.000)
LotConfig__CulDSac	0.032** (0.016)
LotConfig__FR2	-0.025 (0.021)
LotConfig__FR3	-0.077 (0.084)
LotConfig__Inside	-0.003 (0.009)
LotShape__IR2	0.029 (0.021)
LotShape__IR3	0.013 (0.043)
LotShape__Reg	-0.007 (0.008)
LowQualFinSF	0.000 (0.000)

Dependent variable: SalePrice	
MSSubClass	-0.000 (0.000)
MSZoning_FV	0.505*** (0.062)
MSZoning_RH	0.452*** (0.062)
MSZoning_RL	0.465*** (0.054)
MSZoning_RM	0.438*** (0.051)
MiscVal	-0.000 (0.000)
MoSold	-0.000 (0.001)
Neighborhood_Blueste	-0.010 (0.114)
Neighborhood_BrDale	-0.086* (0.051)
Neighborhood_BrkSide	-0.001 (0.044)
Neighborhood_ClearCr	0.010 (0.044)
Neighborhood_CollgCr	-0.034 (0.033)
Neighborhood_Crawfor	0.083** (0.040)
Neighborhood_Edwards	-0.082** (0.037)
Neighborhood_Gilbert	-0.042 (0.036)
Neighborhood_IDOTRR	-0.023 (0.051)
Neighborhood_MeadowV	-0.186*** (0.052)
Neighborhood_Mitchel	-0.063* (0.038)
Neighborhood_NAmes	-0.044 (0.036)
Neighborhood_NPkVill	-0.031 (0.081)
Neighborhood_NWAmes	-0.052 (0.037)
Neighborhood_NoRidge	0.001 (0.039)
Neighborhood_NridgHt	0.048 (0.033)
Neighborhood_OldTown	-0.070 (0.044)

Dependent variable: SalePrice	
Neighborhood_SWISU	-0.008 (0.045)
Neighborhood_Sawyer	-0.044 (0.037)
Neighborhood_SawyerW	-0.028 (0.036)
Neighborhood_Somerst	-0.016 (0.042)
Neighborhood_StoneBr	0.106*** (0.038)
Neighborhood_Timber	-0.017 (0.038)
Neighborhood_Veenker	0.020 (0.051)
OpenPorchSF	0.000*** (0.000)
OverallCond	0.035*** (0.004)
OverallQual	0.042*** (0.005)
PavedDrive_P	0.014 (0.025)
PavedDrive_Y	0.020 (0.016)
PoolArea	0.000 (0.000)
RoofMatl_CompShg	2.822*** (0.148)
RoofMatl_Membran	3.371*** (0.226)
RoofMatl_Metal	3.154*** (0.222)
RoofMatl_Roll	2.896*** (0.183)
RoofMatl_TarGrv	2.941*** (0.181)
RoofMatl_WdShake	2.696*** (0.197)
RoofMatl_WdShngl	2.914*** (0.155)
RoofStyle_Gable	0.069 (0.112)
RoofStyle_Gambrel	0.001 (0.123)
RoofStyle_Hip	0.073 (0.112)
RoofStyle_Mansard	0.098 (0.126)

Dependent variable: SalePrice	
RoofStyle__Shed	0.621*** (0.201)
SaleCondition__AdjLand	0.125* (0.064)
SaleCondition__Alloca	0.082* (0.047)
SaleCondition__Family	0.010 (0.029)
SaleCondition__Normal	0.056*** (0.014)
SaleCondition__Partial	-0.029 (0.067)
SaleType__CWD	0.097* (0.057)
SaleType__Con	0.106 (0.078)
SaleType__ConLD	0.248*** (0.055)
SaleType__ConLI	-0.038 (0.051)
SaleType__ConLw	-0.003 (0.060)
SaleType__New	0.122* (0.070)
SaleType__Oth	0.073 (0.064)
SaleType__WD	-0.013 (0.020)
ScreenPorch	0.000*** (0.000)
Street__Pave	0.136** (0.054)
TotRmsAbvGrd	0.008* (0.005)
TotalBsmtSF	0.000*** (0.000)
Utilities__NoSeWa	-0.204* (0.113)
WoodDeckSF	0.000*** (0.000)
YearBuilt	0.002*** (0.000)
YearRemodAdd	0.001*** (0.000)
YrSold	0.000 (0.002)
intercept	1.144 (5.003)
Observations	1168

Note:

*p<0.1; **p<0.05; ***p<0.01

Figure 6. Decision Tree (Optimal Depth)

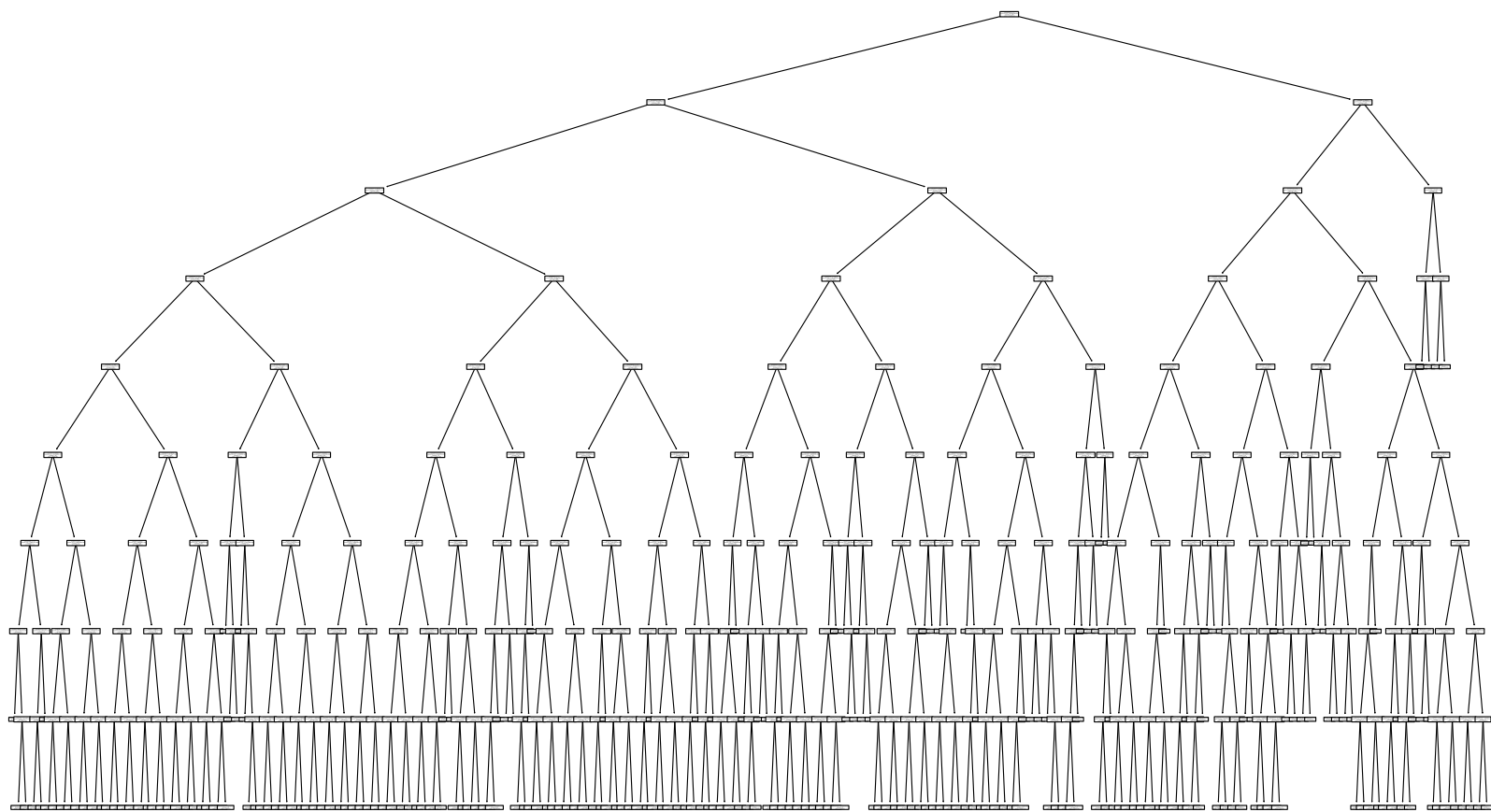


Figure 7. Effect of m on Test MSE in Random Forest